

Automatic time series analysis for electric load forecasting via support vector regression

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HIGHLIGHTS

- A novel strategy for automatic time series forecasting is proposed.
- Support Vector Regression is adapted for feature selection.
- The proposal is applied in energy load forecasting.
- Best predictive performance is achieved in experiments on these datasets.

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ABSTRACT

In this work, a strategy for automatic lag selection in time series analysis is proposed. The method extends the ideas of feature selection with support vector regression, a powerful machine learning tool that can identify nonlinear patterns effectively thanks to the introduction of a kernel function. The proposed approach follows a backward variable elimination procedure based on gradient descent optimisation, iteratively adjusting the widths of an anisotropic Gaussian kernel. Experiments on four electricity demand forecasting datasets demonstrate the virtues of the proposed approach in terms of predictive performance and correct identification of relevant lags and seasonal patterns, compared to well-known strategies for time series analysis designed for energy load forecasting and state-of-the-art strategies for automatic model selection.

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1. Introduction

Forecasting applications across industries observe an increase in time series frequency, from daily retail sales to hourly call centre volumes, half-hourly electricity demand, or minute-by-minute internet traffic load. Such high-frequency time series pose increasing challenges to forecasting methodologies, not only in the volume and velocity of each series, but in the need to model non-linear interactions of exogenous variables, such as price and weather effects, not prevalent in monthly, quarterly, or yearly data. As a result, nonlinear algorithms from machine learning have seen an increasing popularity in forecasting [1], including artificial neural networks [2] and support vector regression (SVR) based on kernel methods [3,4].

In addition, these nonlinear datasets grow big in volume, both from the large number of observations per time series and from the number of time series in a dataset, often extending to thousands of retail products or ten-thousands of electricity meters

with unique time series patterns. These series require a fully automatic specification of the algorithms per each time series, most prominently feature selection of relevant autoregressive lags in interaction with non-linear exogenous variables. However, as complete search spaces of high frequency data include thousands of possible time-lagged features within but one seasonality, wrapper approaches which compute all combinations and select the best candidate ex post become intractable. As an alternative, filter approaches which seek to identify but few suitable candidates prior to their computation including autocorrelation [5] and spectral analysis rely on linear statistics, which are by design inapt to identify the underlying nonlinear input features. On the other hand, SVR cannot automatically identify which variables are most relevant for model estimation. As a result, nonlinear algorithms often employ simple heuristics to specify input features [6], making the model specification of SVR as much an art as a science, thus limiting their automation and usage. As a consequence, applications resort to linear statistical methods of ARIMA and exponential smoothing capable of feature selection, e.g. using Akaike information criterion (AIC) [7].

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Our contribution is twofold. The first is a methodological one since this paper introduces a novel, fully automated methodology for simultaneous input variable or feature selection, model specification and parameter estimation using support vector machines (SVMs) in time series analysis. The proposed method uses a kernel-penalised optimisation of all meta-parameters of SVR, iteratively adjusting the widths of an anisotropic Gaussian kernel for backward variable elimination based on gradient descent optimisation. As such, the algorithm overcomes one of the biggest hindrances in SVR research and application to date, that of identifying nonlinear input features automatically.

The second contribution is an applied one: we establish a relation between research in SVMs and automatic model specification in forecasting. This very interesting challenge was proposed by Hyndman and Khandakar [7] for traditional time series approaches, but this topic has not been studied previously in time series analysis with machine learning methods. In classification tasks, recent innovations in feature selection using support vector machines have been introduced [8], but have not yet been assessed on time series data, where strong serial correlations and multicollinearity between seasonal autoregressive lags void many standard approaches. Experiments on four datasets of electricity demand demonstrate the efficacy of the approach, robustly identifying only relevant lags within complex real-world time patterns of serial correlation, multiple seasonality, outliers, and local time trends, and outperforming established methodologies in automatic model selection and statistical benchmark algorithms in electricity load forecasting.

The paper is organised as follows: Section 2 provides an overview of the SVR method and its extension for feature selection and time series analysis. The proposed framework for automatic time series forecasting is introduced in Section 3, presenting the Kernel-penalised SVR algorithm for automatic lag selection and nonlinear regression. The experimental results are presented in Section 4. Section 5 summarises the key results and discusses directions for future research.

2. Automatic feature selection for SVR

2.1. Previous studies on feature selection for SVR

SVR is a nonlinear algorithm which follows the structural risk minimisation (SRM) principle to obtain excellent generalisation performance [9] instead of minimising the empirical error like other machine learning algorithms, such as artificial neural networks. SVM and SVR are prominent in research and practice, in particular in operational research due to their use of linear optimisation techniques pioneered in operations research to find optimal solutions to nonlinear predictive problems in higher-dimensional feature spaces, see, e.g. [10,11].

In specifying SVMs for classification and regression, feature selection has been identified as an important topic, with growing emphasis on developing novel approaches to handle the challenges derived from big data sets. Maldonado et al. [10] proposed a feature selection method based on SVMs for the churn prediction task. Sanz-Garcia et al. [12] proposed a novel feature selection for SVR based on genetic algorithms, which is applied on predicting temperature settings in a continuous annealing furnace. Ali and Yaman [13] combined feature and instance selection techniques to select a small but informative dataset to train SVR models on six large retail datasets, reducing the number of variables and the multicollinearity and thus increasing the interpretability of the resulting models and their ease of maintenance. Under the different assumption that additional features may be acquired at additional costs, Maldonado and López [8] proposed a novel embedded approach for simultaneous classification and feature selection under class-imbalance condition.

All papers presented innovations to nonlinear feature selection of SVM methodologies in an attempt to facilitate an automatic model specification for different datasets, focussing on specific cases of many features, correlated features or many instances. However, none consider the challenges of autocorrelated features prevalent in time series data, which is understandable considering the nature of classification data. A considerable second group of papers empirically assessed established feature selection methods for SVM, mostly focussing on stock market applications, e.g., Zbikowski [14] using Fisher Score or Song et al. [15] using a genetic algorithm, but without providing an encompassing empirical evaluation. Other papers yet only indicated the importance of feature selection for SVM, but rather focus on other tasks of preprocessing such as variable encoding or scaling [16], and thus added little to the discussion of feature selection in itself.

In contrast, feature selection for SVR in forecasting and time series prediction has received much less attention, often combining existing linear feature selection approaches with nonlinear SVR. Kong et al. [17] proposed a reduced SVR which applies principal component analysis (PCA) for feature selection to forecast wind power plant data, preselecting the PCA factors as a subset of features to identify support vectors in a reduced optimisation problem. However, it remains unclear how a linear PCA can adequately model nonlinear feature interactions in wind speed for any nonlinear SVR to exploit. Similarly, Hu et al. [18] applied SVR for mid-term load forecasting using a (linear) memetic algorithm for feature selection based on the firefly algorithm (MSVR-MA), which aims to select input features among the feature candidates of daily time lagged loads and temperatures on real-world data from North America and Australia. Dai et al. [19] incorporated feature selection into SVR for stock index forecasting using multivariate adaptive regression splines (MARS), a data-driven nonlinear and nonparametric regression methodology which obtains significant predictor variables to serve as the inputs for the SVR model. Although nonlinear in nature, it is unclear how variables considered important by another algorithm can be interpreted by SVR irrespective of the choice of kernel function, which will construct significantly distinct decision boundaries to MARS or other algorithms. Cao and Zhang [20] provided an empirical study of feature selection in SVR for financial time series, investigating two wrapper approaches of saliency analysis (SA), a sensitivity analysis which estimates the importance of features by partial derivatives of the network output on varying a single feature input, and genetic algorithms (GA), a search algorithm mapping all features into a single binary chromosome and randomly perturbing. In addition, a select number of conference papers goes on to combine linear and nonlinear feature selection methods with SVR, yet often with limited empirical evidence and no comprehensive comparison. Son and Kim [21] forecasted short-term electricity demand using SVR with fuzzy-rough feature selection and particle swarm optimisation for 20 variables of monthly electricity consumption, weather and social variables. Sarhani and Afia [11] proposed simultaneous feature selection and parameter optimisation using particle swarm optimisation (PSO) of SVR for an application in electric load forecasting, but remain unclear on the trade-off of PSO optimised support vectors versus the conventional linear optimisation in higher dimensional space, thus questioning the very use of SVR.

While all authors found that feature selection in general improves forecasting performance of SVR, no single paper proposed a novel approach that is not disjunct from the SVR methodology, be it linear in nature or a wrapper estimating thousands of potential candidate models. Thus, no feature selection routine for SVR draws upon the inherent strengths of linear optimisation in higher dimensional feature space, lacking integration and thus automation and suitability.

It should be noted that this omission is not unique to forecasting with SVR, but similar gaps persist for other machine learning algorithms, most notably artificial neural networks, see, e.g., Crone and Kourentzes [6], Guajardo et al. [22]. The authors speculate whether this lack of integrated feature selection for SVR and ANN have contributed to the prominence of decision trees and ensemble-variants thereof, most notably random forests and gradient boosting, which implicitly conduct feature selection in growing trees of limited complexity.

2.2. SVR for time series analysis

This section presents ϵ -SVR [23,24] for time series analysis. Given a set of time series data, a training set consisting of a continuous dependent variable $y_i \in \mathbb{R} \forall i = 1, \dots, m$ and covariates $\mathbf{x}_i \in \mathbb{R}^p, i = 1, \dots, m$ can be constructed by taking p lags of y_i . The method ϵ -SVR constructs a function $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$, $\mathbf{w} \in \mathbb{R}^n, b \in \mathbb{R}$ that has at most a deviation of ϵ from y_i for all training instances $\mathbf{x}_i$, and at the same time is as flat as possible [24]. Flatness is encouraged by minimising the Euclidean norm of $\mathbf{w}$, while model fit is achieved by penalising the sum of the deviations higher than ϵ . The ϵ -SVR method aims at solving the following optimisation problem:

$$\begin{aligned} \min_{\mathbf{w}, b, \xi, \xi^*} \quad & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^m (\xi_i + \xi_i^*) \\ \text{s.t.} \quad & y_i - (\mathbf{w}^T \mathbf{x}_i + b) \geq \epsilon + \xi_i, \quad i = 1, \dots, m, \\ & (\mathbf{w}^T \mathbf{x}_i + b) - y_i \geq \epsilon + \xi_i^*, \quad i = 1, \dots, m, \\ & \xi_i, \xi_i^* \geq 0, \quad i = 1, \dots, m, \end{aligned} \quad (1)$$

where the slack variables $\xi, \xi^* \in \mathbb{R}^m$ are introduced for each training vector in order to allow deviations higher than ϵ , but penalising these deviations in the objective function. The parameter C controls the trade-off between model fit and complexity reduction [24].

Non-linear model estimation can be obtained by mapping the data to a higher dimensional feature space $\mathcal{H}$. Instead of using a projection function, Eq. (1) can be represented in its dual form, where the data points appear only in the form of dot products. The mapping is performed by a kernel function $K(\mathbf{x}_i, \mathbf{x}_s)$ which defines an inner product in $\mathcal{H}$ [25]. Once applying the KKT conditions, the following dual formulation is obtained:

$$\begin{aligned} \max_{\alpha, \alpha^*} \quad & \sum_{i=1}^m y_i (\alpha_i - \alpha_i^*) - \epsilon \sum_{i=1}^m (\alpha_i + \alpha_i^*) \\ & - \frac{1}{2} \sum_{i,s=1}^m (\alpha_i - \alpha_i^*) (\alpha_s - \alpha_s^*) K(\mathbf{x}_i, \mathbf{x}_s) \\ \text{s.t.} \quad & \sum_{i=1}^m (\alpha_i - \alpha_i^*) = 0, \\ & 0 \leq \alpha_i \leq C, \quad i = 1, \dots, m, \\ & 0 \leq \alpha_i^* \leq C, \quad i = 1, \dots, m. \end{aligned} \quad (2)$$

The decision rule $f(\mathbf{x})$ for a new sample $\mathbf{x}$ is thus:

$$f(\mathbf{x}) = \sum_{i=1}^m (\alpha_i - \alpha_i^*) K(\mathbf{x}, \mathbf{x}_i) + b \quad (3)$$

Among a variety of kernel functions available, the Gaussian kernel is chosen in many applications due to its superior performance [26]. This function has the following form: $K(\mathbf{x}_i, \mathbf{x}_s) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_s\|^2}{2\sigma^2}\right)$, where $\sigma > 0$ is the parameter controlling the width of the kernel.

3. Kernel-penalised SVR for automatic feature selection

We propose an extension of kernel-based SVR (Eq. (2)) that automatically selects the lags that are relevant in the construction of the regression function. The main idea is to improve the performance of SVR in time series by removing the lags that are irrelevant for the solution. The proposal optimises a vector of scaling factors incorporated in SVR, whose cardinality is penalised using a zero-norm approximation. Then, the lags are excluded in a backward elimination strategy. This strategy has the advantage of performing automatic lag selection by taking into account the interactions between the lags and the regression model.

We limit ourselves to Gaussian kernel functions, and we use the concept of the anisotropic Gaussian kernel [27], which has the following form:

$$K(\sigma * \mathbf{x}_i, \sigma * \mathbf{x}_s) = \exp\left(-\frac{\|\sigma * \mathbf{x}_i - \sigma * \mathbf{x}_s\|^2}{2}\right) \quad (4)$$

where σ are the widths of an anisotropic Gaussian kernel and $*$ is the componentwise vector product operator. The main idea is to update vector σ iteratively via successive gradient steps, similarly to the scaling factors technique [28,29]. Originally developed for the standard soft-margin SVM classification [27], the KP-SVM method was extended to regression in [3], and it is subsequently studied for time series analysis in the present work.

A penalisation function for the use of covariates is included in the algorithm in order to minimise the non-zero elements in σ . The cardinality of the vector can be approximated by the following function [30]:

$$\text{Card}(\sigma) \approx \sum_{j=1}^p [1 - \exp(-\beta \sigma_j)], \quad (5)$$

where $\beta = 5$ was suggested in Bradley and Mangasarian [30]. We propose a two-step strategy to optimise both $\{\alpha, \alpha^*\}$ and σ :

- We train SVR for a fixed width σ^0 , obtaining the solution tuple $\{\alpha^0, \alpha^{*0}\}$ that leads to regression function. The first formulation follows:

$$\begin{aligned} \max_{\alpha, \alpha^*} \quad & \sum_{i=1}^m y_i (\alpha_i - \alpha_i^*) - \epsilon \sum_{i=1}^m (\alpha_i + \alpha_i^*) \\ & - \frac{1}{2} \sum_{i,s=1}^m (\alpha_i - \alpha_i^*) (\alpha_s - \alpha_s^*) K(\sigma^0 * \mathbf{x}_i, \sigma^0 * \mathbf{x}_s) \\ \text{s.t.} \quad & \sum_{i=1}^m (\alpha_i - \alpha_i^*) = 0, \\ & 0 \leq \alpha_i \leq C, \quad i = 1, \dots, m, \\ & 0 \leq \alpha_i^* \leq C, \quad i = 1, \dots, m. \end{aligned} \quad (6)$$

- For a fixed solution tuple $\{\alpha^0, \alpha^{*0}\}$, we aim at penalising the use of covariates via the following nonlinear formulation:

$$\begin{aligned} \min_{\sigma} \quad & F(\sigma) = \sum_{i,s=1}^m (\alpha_i^0 - \alpha_i^{*0}) (\alpha_s^0 - \alpha_s^{*0}) K(\sigma * \mathbf{x}_i, \sigma * \mathbf{x}_s) \\ & + C_2 \sum_{j=1}^p [1 - \exp(-\beta \sigma_j)] \\ \text{s.t.} \quad & \sigma_j \geq 0 \quad j = 1, \dots, p. \end{aligned} \quad (7)$$

Notice that if we want to fix the α values, then all constraints from SVR are constant since the data points only appear in the objective function in the form of a kernel function, leading to a

unconstrained problem. We propose solving Eq. (7) via a gradient descent strategy, removing those variables whose respective widths are close to zero.

Following the notation used by Song et al. [15], the set of available covariates is denoted by $\mathcal{S}$, initialised as the full set of features. The subset of regressors to be removed from $\mathcal{S}$ at each iteration is denoted by $\mathcal{I}$. The following algorithm is proposed for this backward elimination strategy:

Algorithm 1 Kernel-penalised SVR for time-series analysis

1. $\sigma \leftarrow \sigma_0 \mathbf{e}$
 2. **repeat**
 3. $\{\alpha^o, \alpha^{*o}\} \leftarrow$ SVR Training on $\mathcal{S}$ (Eq. (6))
 4. **repeat (step 2)**
 5. $\sigma \leftarrow \sigma + \lambda \nabla F(\sigma)$
 6. **until** $\exists j \mid \sigma_j < \epsilon^*$
 7. $\mathcal{I} \leftarrow \{j \in \mathcal{S} \mid \sigma_j < \epsilon^*\}$
 8. $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}$
 9. **until** Convergence is reached or $|\mathcal{S}| = r$
-

The step 2 of the algorithm iteratively updates σ by using successive gradient descent steps, where λ is a predefined gradient parameter and $\nabla F(\sigma)$ is the gradient of Eq. (7) (see line 5 of Algorithm 1). The partial derivative of $F(\sigma)$ with respect to σ_j follows:

$$\begin{aligned} \frac{\partial F(\sigma)}{\partial \sigma_j} = & C_2 \beta \exp(-\beta \sigma_j) \\ & - \frac{1}{2} \sum_{i,s=1}^m \sigma_j (x_{ij} - x_{sj})^2 (\alpha_i^o - \alpha_i^{*o}) \\ & \times (\alpha_s^o - \alpha_s^{*o}) K(\sigma * \mathbf{x}_i, \sigma * \mathbf{x}_s). \end{aligned} \quad (8)$$

Initialised as a standard isotropic Gaussian kernel (line 1 of Algorithm 1), the components of σ should remain positive, and therefore we remove the variables in $\mathcal{I}$ such that $\sigma_j < \epsilon^*$ and return to step 1, where ϵ^* is a predefined threshold (see lines 6–8 of Algorithm 1).

Finally, the algorithm stops when convergence is reached, i.e. the scaling factors do not change significantly between iterations, which can be monitored by computing the one-norm of σ and setting a predefined threshold. The algorithm can also stop when a predefined number of desired covariates r is found (see line 9 of Algorithm 1).

To extend the KP-SVR method to time series we adapted the SVR projection of points to intervals ordered in time, by feeding back the firsts projection as a lag for the next forecasts.

As an embedded feature selection method, KP-SVR has the advantage of taking into account the interaction between variables, being usually less time-consuming than wrapper methods, such as Auto-ARIMA. SVR also has the flexibility of constructing nonlinear models thanks to the use of kernel functions, and does not make assumptions on the data. However, the main limitation of KP-SVR is that many parameters need to be determined, leading to a potentially time-consuming validation process for model selection.

4. Experimental design

4.1. Datasets in load forecasting

A significant number of papers on SVR, and feature selection in particular, have considered electricity load data. Not only

does load forecasting remain essential for all managerial decisions where the operational costs are affected by forecasting errors [31], from strategic infrastructure development of networks and metres, to short-term operational decisions of electricity generation for the next 24 to 36 h ahead, where an increase of 1% forecast error can induce a US\$ 10 million increase in operational costs [32]. More importantly, accurate load forecasting remains a challenge due to its underlying data generating processes: electricity load is a random non-stationary process [33] comprised of a triple seasonality (daily, weekly and annually) [32] and implicit non-linearity of exogenous variables [34,35], with frequent outliers, outlier patches, local time trends and a mixture of deterministic and stochastic seasonal patterns. Additionally, the relevant features may depend on the planning horizon, which is typically differentiated as short-term load forecasting (for up to one day), medium-term load forecasting (one day to one year), and long-term load forecasting (one to ten years) [36]. As a result, SVR has been widely studied in load forecasting, see e.g. the survey by Sapankevych and Sankar [4]. Whilst the majority of publications propose different methodologies to determine the free parameters of ϵ -SVR (C, ϵ, σ) [35,37–39], selected paper including Dong et al. [40] incorporate nonlinear exogenous weather factors such as temperature, humidity, weather conditions and solar radiation, in addition to autoregressive lags, exacerbating the challenges of feature selection. As the performance of SVR in load forecasting depends on an adequate feature selection [3], we consider this a valid empirical application area for our contribution in SVR feature selection.

We applied the proposed methodology to four datasets (the Great Britain -GB- load forecasting, the France load forecasting, the GB 'IO14_DEM' data, and the GB 'IO14_TGSD' data) following two different experiments: First, Experiment A, we studied high-frequency data by performing half-hourly forecast, taking the autoregressive process into account and including daily and weekly seasonality. Second, Experiment B, a daily forecast was performed for the Great Britain dataset, and two exogenous variables were included: the forecasted temperature for the given day, and a dummy variable that represents whether or not a given day is a holiday. The temperature is known to have a nonlinear relationship with respect to energy consumption [41], allowing, therefore, an interesting experiment for assessing the virtues of a kernel-based approach such as SVR.

The Great Britain and France load datasets are similar compared to the ones used by Taylor [5] but are of more recent years. The 'IO14_DEM' and 'IO14_TGSD' energy datasets were used in Hawkins et al. [42] and Bejan et al. [43], respectively. For all the datasets in experiment A, we considered 14 months of half-hourly observations from September 2007 to October 2008 (20,496 samples). For experiment B, we selected seven years and eight months of daily electricity demand, from March 2001 to October 2008.

4.2. KP-SVR specification

Following the suggestions in Hong [44] the data set was split into three subsets: training set (Exp. A: September 2007 to August 2008, and Exp. B: March 2001 to 3rd of November 2006), validation set (Exp. A: September 2008, Exp. B: 4th November 2006 to 11th November 2007), and test set (Exp. A: October 2008, Exp. B: 3rd November 2007 to 31 October 2008). All model parameters were estimated on the training and validation set, using the rolling forecasting strategy [45], while the final model assessment was performed over the test set that remained unused during parameterisation.

The rolling forecasting strategy adapts crossvalidation to time series data. Traditional crossvalidation splits the data randomly

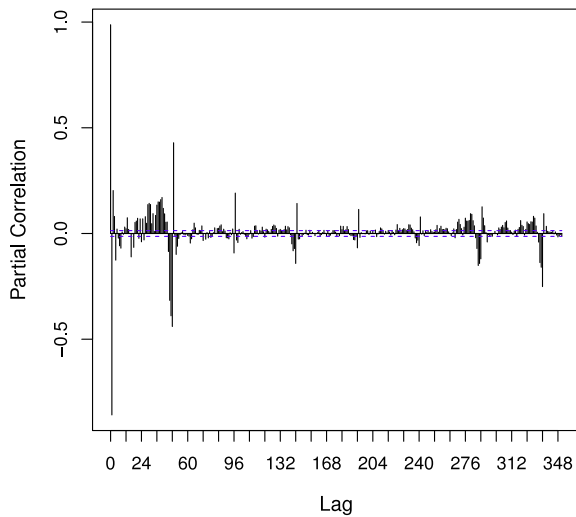


Fig. 1. PACF plot for the first 352 lags.

into a training set, in which the model is fitted, and a validation set, in which the accuracy of the fitted model is assessed. The process is usually repeated several times varying both subsets, and the performance of the method is finally computed as the average of the performance of all the validation sets. For time series analysis, however, the training set must include only observations that were made prior to the ones that constitute the test set [45].

In experiment A, we always train our model using a full year of data and test it using the subsequent day (48 half-hourly observations). The origin is then “rolled” one day forward, and the process is repeated. The average of a whole month is computed (September 2008 and October 2008 for model selection and final testing, respectively), and the Mean Absolute Percentage Error (MAPE) is used as the main performance metric. The same process is executed for experiment B, testing and “rolling” four weeks (28 daily observations).

The feature construction process was a challenging task due to the size of the dataset. For the half-hourly prediction, the first 352 lags were selected as candidate variables for taking daily and weekly seasonality into account [5,32]. For the daily forecast dataset, the first 380 lags were selected as candidate variables, including weekly and yearly seasonality. For the latter set of experiments, all variables were scaled between zero and one to guarantee that both temperature and energy consumption would have the same scale.

The PACF (Partial Auto-Correlation Function) of the GB load forecasting dataset, which can be seen as a measure of relevance for model selection, is presented next for illustrative purposes. PACF values that are significantly different from zero are usually pre-selected, resulting in a filter method for feature selection. Although this strategy can be useful in traditional time series datasets, the main issue with high frequency data is that most lags are significantly different from zero, and therefore few attributes are filtered out with this strategy. The PACF plot is presented in Fig. 1, which includes the thresholds for statistical significance (the blue dashed lines).

In Fig. 1, we clearly observe the described phenomena: according to the PACF strategy, most lags must be included in the modelling process, since it is unable to provide a solution with few covariates. Another expected result is that the highest partial auto-correlations are the first ones, around lag 48 (daily seasonality) and lag 336 (weekly seasonality).

4.3. Benchmark algorithms

We will assess the proposed feature selection using kernel penalised SVR on electric load data, in comparison to the established algorithms of load forecasting [44,46]: statistical methods of SARIMA and trend-seasonal exponential smoothing, see, e.g., Mbamalu and El-Hawary [47] for autoregressive models with seasonal factors or Taylor [5] for exponential smoothing methods up to the triple seasonality respectively, and artificial neural networks, see, e.g. Darbellay and Slama [2]. These methods typically rely on feature and model selection based on information theoretic performance metrics. Most notably, Hyndman and Khandakar [7] proposed ARIMA and exponential smoothing extensions that automatically select the relevant lags in time series analysis by using the Akaike’s Information Criterion (AIC), which serve as strong benchmark methods. Finally, we implemented artificial neuronal networks (ANNs) for time series analysis without performing feature selection and considering the same input variables as SVR and KP-SVR. We consider 100 nodes in a single hidden layer, the rectified linear unit (ReLU) activation function, and 500 epochs.

Regarding model implementation, we used LIBSVM [48] on Matlab for the SVR approaches, the forecast package implemented in R for the SARIMA, double seasonal Holt-Winters, and Auto-ARIMA approaches [7], and the MLPRegressor function in Python for the ANN method.

4.4. Experiment A on half-hourly data

The methodological procedure for SVR without covariate selection and the proposed approach follows:

- We first trained SVR using all the available covariates. In this step we identified the best configuration for a set of possible hyperparameters. The following values were studied: $\epsilon \in \{0.01, 0.1, 0.5, 1\}$, $C, \sigma \in \{2^{-15}, 2^{-14}, \dots, 2^{14}, 2^{15}\}$.
- For model selection we followed the rolling forecast validation procedure described previously. For the best model selected, covariate selection was performed on the training set of the rolling forecast procedure via the Kernel-penalised algorithm.
- For the Kernel-penalised SVR approach, we set $\epsilon^* = 0.000001$, $\lambda = 0.00001$, and $C_2 = 1$. The threshold parameter ϵ^* has to be sufficiently small in order to avoid the removal of relevant features, while λ has to provide a good compromise between σ and the gradient of $F(\sigma)$ in order to avoid a strong fluctuation of the kernel variables and negative widths, especially at the first iterations of the algorithm. Finally, setting $C_2 = 1$ provides a balanced trade-off between feature selection and margin maximisation in $F(\sigma)$.

The best SVR and KP-SVR models obtained using the **validation** set were compared with the remaining benchmark approaches using the **test** set. Table 1 summarises the test performance for each method and for all datasets in terms of MAPE. The best performance (lowest MAPE) is highlighted in bold type.

In Table 1 we observe that KP-SVR achieves the best performance in terms of MAPE in three of the four datasets, demonstrating the virtues of our approach in finding the most adequate set of variables and constructing effective models simultaneously. SVR using all variables, and double seasonal Holt-Winters are slightly above our proposal in terms of MAPE, achieving remarkably better results compared with the remaining methods. Since these other methods only take into account one seasonal pattern (SAR, SMA, SARMA, Auto-ARIMA), we confirm the importance of incorporating daily and weekly seasonal patterns in the modelling

Table 1

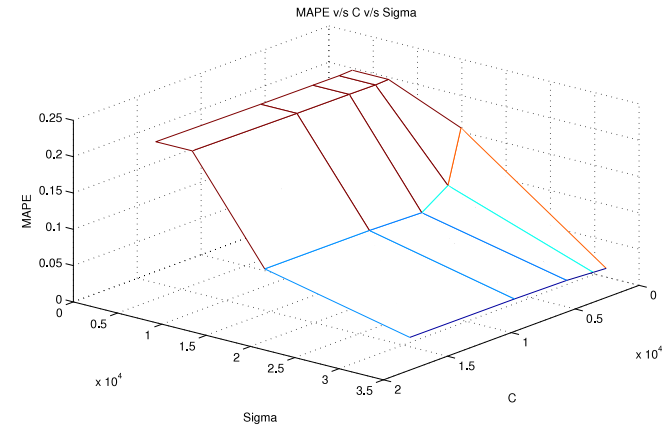
Test performance summary for different time series approaches.

	DS-HW	SAR	SMA	SARMA	Auto-ARIMA	ANN	SVR	KP-SVR
GB	2.68	4.82	10.18	5.95	24.76	2.70	2.24	1.96
France	3.17	4.86	10.38	5.93	5.80	5.48	2.12	2.13
IO14_DEM	1.96	4.73	9.98	5.83	28.52	4.97	2.17	1.92
IO14_TGSD	2.34	4.65	8.67	5.54	5.28	4.30	2.33	2.05

Table 2

Running times, in minutes, for all datasets and methods.

	DS-HW	SAR	SMA	SARMA	Auto-ARIMA	ANN	SVR	KP-SVR
GB	2'.569	0'.291	0'.177	0'.688	11'.94	0'.303	101'.72	166'.62
France	2'.667	0'.457	0'.374	1'.509	222'.42	0'.078	125'.47	193'.48
IO14_DEM	2'.445	0'.618	0'.346	1'.632	99'.20	0'.113	96'.36	119'.71
IO14_TGSD	3'.372	1'.010	0'.294	1'.181	117'.81	0'.242	82'.09	201'.77

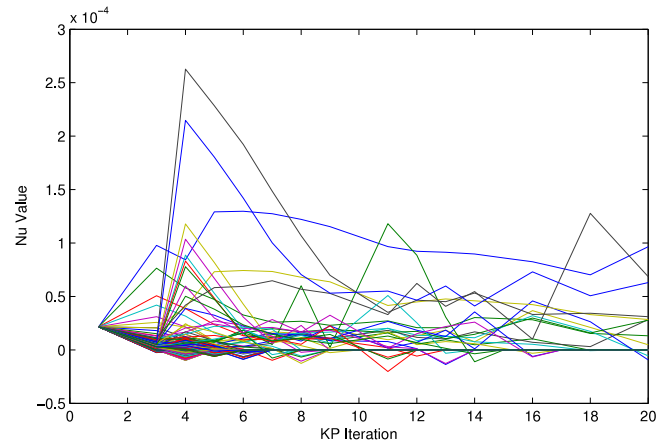
**Fig. 2.** Model selection procedure for SVR using 352 attributes.

process. Although the ANN method considers both seasonal patterns, it is not able to achieve the positive results obtained by SVR.

The running times, in minutes, for all methods and datasets are presented in Table 2. These experiments were performed on an HP Envy dv6 with 16 GB RAM (750 GB SSD) with Microsoft Windows 8.1 Operating System (64-bits), and an i7-2620M processor with 2.70 GHz.

It can be seen on Table 2 that all running times are tractable and under four hours for all methods and datasets. As expected, our proposed method is slower than SVR since it requires an initial model training with all the covariates considered, i.e. KP-SVR contains the SVR method. However, our proposal requires relatively few iterations of small-sized SVR problems, and therefore requires 172% of time in average when compared with SVR. On the other hand, Auto-ARIMA is very slow and requires more than eighty times more time in average when compared to standard ARIMA. It is also only tractable because it explores seasonal patterns up to 48 lags, in contrast with KP-SVR that explores 350 lags. The exhaustive search performed by Auto-ARIMA is intractable for our problem. Finally, the Python implementation for the ANN method is extremely efficient, achieving similar training times when compared to the ARIMA methods.

A series of experiments using the Great Britain load forecasting dataset follow for illustrative purposes. First, the results for SVR using the first 352 lags are reported for different values of C and σ . We observed stable results for parameter ϵ , which we fixed in $\epsilon = 0.1$. The MAPE obtained for these experiments is presented in Fig. 2.

**Fig. 3.** KP-SVR method. Evolution of kernel width σ in each iteration.

In Fig. 2 we observe that the best configuration is obtained for high values of C and σ ($C = 2^{11}$ and $\sigma = 2^{15}$), being relatively stable for parameter C , but showing a high variance for the kernel width. We conclude that an adequate grid search is highly recommended in order to guarantee adequate predictive results, using a comprehensible set of possible values for all parameters.

Next, the performance of the proposed KP-SVR approach for time series is studied. First, the evolution of the anisotropic kernel width parameter σ is presented from the isotropic value obtained in the model selection step (iteration 1) until the final solution is obtained (iteration 20). The stopping criterion is obtained based on the performance in the validation set: the algorithm stops when the removal of attributes implies a significant decrease in performance. Fig. 3 presents the values for each of the components of σ (Y-axis) at each iteration (X-axis).

In Fig. 3 we observe that most of the 350 lines are available during the first 4–8 iterations, observing a dense zone of lines in the left half of the figure. However, only few lines are available after 10 iterations (right half of the figure), meaning that most nu values are set to zero since they fall below the threshold $\epsilon*$. Interestingly, some σ values increase with respect to the isotropic kernel width, while others are below this value. The final solution has the following nine lags: 1, 2, 6, 14, 44, 48, 49, 336, and 337. According to these values, we conclude that the proposed method is able to successfully identify the lags that are relevant for short-term electric load forecasting, which are the first two lags, daily seasonality (around lag 48), and weekly seasonality (at around lag 336). Notice that the algorithm received the first 352 lags without any additional information of seasonality; it found that the seasonal patterns were relevant for this problem automatically.

Next, the validation performance of KP-SVR in terms of MAPE is reported for each iteration. The Y-axis presents the MAPE, while the X-axis presents the number of selected variables (lags) in each iteration. The dots correspond to the performance of KP-SVR from the first iteration (number of lags=352) to the final solution obtained in iteration 20 (the red dot). We also include additional iterations with fewer variables for illustrative purposes. Notice that this plot should be read backwards since the greater the number of iterations, the fewer the variables included in the solution. Fig. 4 presents the full validation performance, while Fig. 5 represents a left section of Fig. 4, in which we focused on the last iterations for illustrative purposes. The red dashed line corresponds to the validation performance of SVR with all variables (MAPE = 2.66%), which is also the performance of KP-SVR in the first iteration.

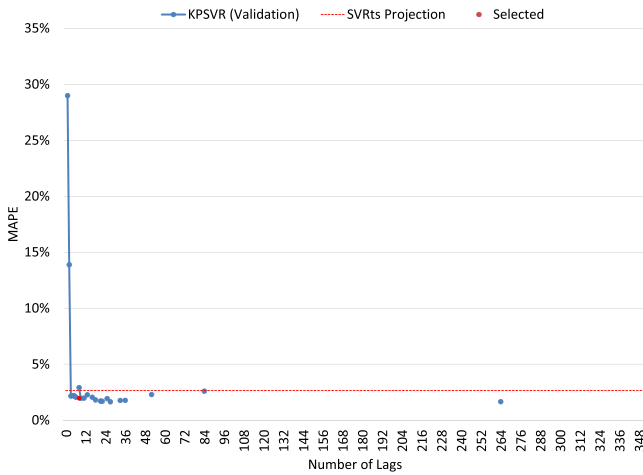


Fig. 4. Feature selection performance for KP-SVR.

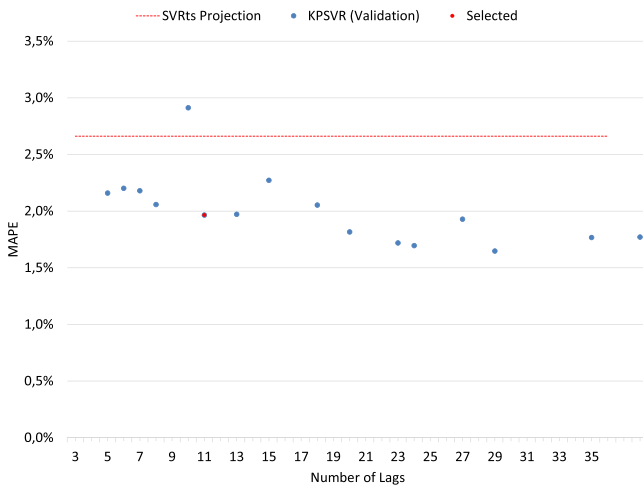


Fig. 5. Feature selection performance for KP-SVR (section).

In Figs. 4 and 5 we observe that KP-SVR performs better than SVR in general, achieving a MAPE of 1.96% for the final solution (the red dot, with nine attributes). In the next iteration, the removal of one attribute leads to a significant loss in terms of performance, and therefore the model obtained with nine attributes was therefore selected as the final model.

Fig. 6 presents the results for the stability analysis. The blue bars represent the percentage of times a given lag (X-axis) was selected among the nine most relevant for KP-SVR. The red bars represent the absolute values for the PACF, also in percentages (100% corresponds to a partial auto-correlation of 1 or -1).

In Fig. 6 we first observe that the KP-SVR is consistent at identifying similar sets of relevant variables since the percentage of times a variable is selected is either zero or close to 100 for most lags. Additionally, it can be seen clearly that the KP-SVR method consistently selects the lags that are relevant a priori: lags 1 and 2, lags 48 and 49 (daily seasonality), and lags 336 and 337 (weekly seasonality). Lags that are unrelated to this problem a priori, such as the variables between lag 6 and 22, appear only in some cases.

4.5. Experiment B on daily data with exogenous variables

The following set of experiments consider the inclusion of two exogenous variables: the average temperature forecast for the

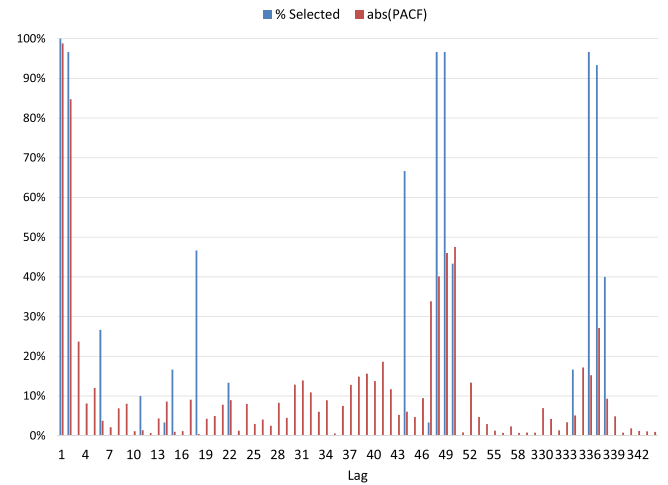


Fig. 6. Stability analysis for KP-SVR and the relationship with PACF.

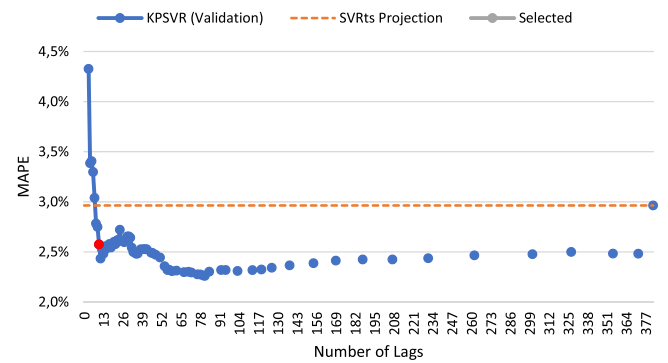


Fig. 7. Feature selection performance for KP-SVR. Daily forecast with exogenous variables.

corresponding day, and whether the day is a bank holiday. The same methodological procedure executed at experiment A was used for experiment B.

First, the validation performance of the KP-SVR method is presented for an increasing number of selected variables, including the two exogenous variables and the autoregressive process with two seasonal patterns (weekly and yearly seasonality). Similar to Figs. 4 and 5, the MAPE of the solution is presented on the Y-axis, each dot representing a different KP-SVR solution, and the dashed line showing the validation performance for SVR including all the variables.

In Fig. 7, it can be observed that KP-SVR achieves a better performance from the beginning of the backward elimination process until approximately 11 attributes are selected; the removal of the subsequent attributes leads to an important increase in the validation error. The final solution has the following 11 variables: 1, 2, 3, 7, 52, 112, 350, 364, 365, 367, and the temperature forecast. It can be concluded that the proposal identifies the seasonal patterns that are relevant for this task successfully, which are the first two-three lags, weekly seasonality (lag 7), and yearly seasonality (around lag 365). The method considers the temperature as relevant for the task, but it eliminates the information related to the bank holidays (this variable is ranked 17).

Next, the test performance for all the methods is presented in Table 3 for the daily forecast dataset, similarly to the results presented in Table 1. The best performance (lowest MAPE) is highlighted in bold type. Results without the exogenous variables

Table 3

Test performance summary for different time series approaches. Daily forecast with and without exogenous variables.

	DS-HW	SAR	SMA	SARMA	Auto-ARIMA	ANN	SVR	KP-SVR
MAPE ^a	4.87	6.11	9.9	6.08	7.63	4.90	2.66	2.74
MAPE	–	5.02	7.28	5.92	6.55	5.01	2.24	1.79

^aThis set of experiments does not include exogenous variables.

are also included for completeness. The method double-seasonal Holt-Winters is reported only for the set of experiments that does not include exogenous variables since this method has not yet been formalised for this task, to the best of our knowledge.

In Table 3, it can be seen that KP-SVR achieves the best performance for both error measures when the exogenous variables are included. This demonstrates the virtues of our approach in finding the most adequate set of predictors while constructing the regression function. The use of kernel functions allows the approach to capture the nonlinear relationship between energy consumption and temperature adequately. This is reflected by the important gain in performance that results from using SVR and KP-SVR when compared with the alternative approaches. Also notable is the improvement that represents the exogenous variables in the daily forecast in comparison with the models that do not include this information (second and first rows in Table 3, respectively). It can be concluded that the inclusion of the two seasonal patterns and exogenous variables, such as the temperature, are crucial for successfully predicting the daily energy load, while the kernel methods outperform traditional forecasting techniques such as Holt-Winters or ARIMA. Our approach is flexible enough to incorporate all these features and to find the best model automatically.

5. Conclusions

An automatic model selection strategy for time series analysis using SVR is presented in this work. An embedded feature selection method is proposed to provide a framework in which the relevant lags are identified simultaneously with the construction of the regression model. The method also has the advantages of SVR: it is able to construct nonlinear regressors thanks to the use of kernel functions and is also less prone to overfitting compared to neural networks given the ϵ -insensitive loss function used to maximise model fit. SVR is also more robust than neural networks since it constructs a convex optimisation problem, leading to a single optimal solution.

The proposed framework was applied to four short-term energy load forecasting datasets. This task requires advanced tools in order to identify the relevant lags and the seasonal patterns automatically given its high-frequency nature. Methods that perform an exhaustive search quickly become intractable since feature selection is a combinatorial problem, and the cost of those algorithms increases exponentially with the number of variables pre-selected.

Our main contribution is setting a parallel between embedded feature selection and automatic time series analysis. The former field has delivered a plethora of efficient methods that can be used in energy load forecasting and time series analysis in general. In particular, the KP-SVR method was used in this work successfully, leading to the following results:

- It performed better than other benchmark approaches in terms of MAPE, thanks to its ability to construct a flexible regressor using few variables.
- It was able to identify the lags that are relevant a priori, such as the first two lags and the two seasonal patterns.

- It discarded irrelevant variables efficiently since it required only 20–30 iterations to construct the final model with only 9 to 11 variables. Exhaustive search algorithms are bottlenecked due to the high complexity of the problem, and therefore heuristic approaches are more suitable for high-frequency data.
- It provided stable solutions, identifying similar sets of relevant lags while varying the training set using the rolling forecast origin strategy.
- It modelled nonlinear relationships successfully between exogenous covariates and the target variable.

Some further research opportunities were identified in this work. Backward feature selection approaches are more time-consuming than forward strategies, and the latter methods are more suitable for high-frequency data. In particular, our study explores the first two seasonal patterns for this reason, but we expect to be able to identify the yearly seasonality effectively with embedded methods for SVR using forward feature selection. Additionally, an interesting line of research that can be derived from this work is the use of efficient strategies for SVM optimisation in time series analysis. Highly-optimised implementations designed to achieve reduced training times have already been proposed for SVM classification (see, e.g. [49]), and such strategies can be adapted first to SVR, and subsequently to time series analysis in order to construct robust regressors efficiently using all available information when facing high-frequency data.

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Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to <https://doi.org/10.1016/j.asoc.2019.105616>.

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